

大模型控制小车

前置说明

```
# 首先开发环境包括虚拟机下的Ubuntu20.04系统（x86_64）以及rockpi 5b的Ubuntu20.04系统（ARM）；为避免用户混淆概念，我们在指令前缀添加系统标识用以区分

# windows下安装的虚拟机中的Ubuntu20.04系统（x86_64）
Linux $ ...

# rockpi 5b下的Ubuntu20.04系统（ARM）
rock $ ...
```

准备工作

复制脚本到指定目录

```
rock $ cd ~/Desktop/ai_app/RKSDK
rock $ cp car_cmd.sh ~/Desktop/rock_ws/ros_ws/
rock $ cp car_cmd_daemon.py ~/Desktop/rock_ws/ros_ws/
```

启动ros

终端1

```
rock $ cd ~/Desktop/rock_ws/microros_ws
rock $ source install/setup.bash
rock $ ros2 run micro_ros_agent micro_ros_agent udp4 --port 8888
```

终端2

```
# 若 /dev/rt_shell 不存在，先跑一次（会创建软链接）
rock $ bash ~/Desktop/rock_ws/ros_ws/usb_auto_setup.sh

rock $ minicom -D /dev/rt_shell -b 1500000
```

进入 RT-Thread shell 后执行：

```
rock $ microros_chassis udp 10.10.10.31 8888
rock $ chassis_car_app
```

启动大模型服务

终端3

```
rock $ cd ~/Desktop/ai_app/RKSDK/test_rkllm_run
rock $ source ../../.venv/bin/activate
rock $ python flask_server.py
```

```
● rock@rock-5b:~/Desktop/ai_app/RKSDK/test_rkllm_run$ source ../../.venv/bin/activate
○ (.venv) rock@rock-5b:~/Desktop/ai_app/RKSDK/test_rkllm_run$ python flask_server.py
[sudo] password for rock:
bash: fix_freq_rk3588.sh: No such file or directory
=====init.....=====
I rkllm: rkllm-runtime version: 1.2.3, rknpu driver version: 0.9.8, platform: RK3588
I rkllm: loading rkllm model from models/_W8A8_RK3588.rkllm
I rkllm: rkllm-toolkit version: 1.2.3, max_context_limit: 4096, npu_core_num: 3, target_platform: RK3588, model_dtype: W8A8
I rkllm: Enabled cpus: [0, 1, 2, 3]
I rkllm: Enabled cpus num: 4

rkllm init success!

=====
* Serving Flask app 'flask_server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8001
* Running on http://192.168.137.99:8001
Press CTRL+C to quit
```

终端4

aichat.py 可实现对话控制大模型移动，并且可自由修改小车移动的线速度和角速度。

当前默认的距离和角度换算为：

- 前进/后退 1 米对应 1 秒
- 左/右转 90 度对应 4 秒

```
rock $ cd ~/Desktop/ai_app/RKSDK/test_rkllm_run
rock $ source ../../.venv/bin/activate
rock $ python aichat.py --yes
```

```
○ (.venv) rock@rock-5b:~/Desktop/ai_app/RKSDK/test_rkllm_run$ python aichat.py --yes
输入 /exit 退出；输入 /stop 紧急停止；输入 /warmup 预热；输入 /status 查看后台发布者状态。
You: 前进2米，左转30度
Assistant: <rtt_0>(2)
<rtt_2>(30)
<|im_end|>
执行计划：
  1. 前进 speed=0.150 duration=2.00s
  2. 左转 speed=0.300 duration=1.33s
WARMED topic=/cmd_vel qos=best_effort rate=20.0
action=fwd speed=0.15 duration=2.0 topic=/cmd_vel qos=best_effort
action=ccw speed=0.3 duration=1.3333333333333335 topic=/cmd_vel qos=best_effort
```